

Varad Nevasekar

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Education

University of Massachusetts Lowell

National Scholarship Awardee / Double River Hawk Graduate Fellowship

Ph.D. Computer Science

May 2029 | GPA: 4.0

M.S. Computer Science

December 2024 | GPA: 4.0

B.S. Computer Science

December 2023 | GPA: 4.0

- Chancellor's Medal for Distinguished Academic Achievement

Publications

V. Nevasekar, M. E. Cabrera, "Seeing Eye Stretch: Robot-assisted Indoor Navigation for Blind or Low-Vision Individuals," *IEEE Ubiquitous Robots (UR)*, 2025, doi: 10.1109/UR65550.2025.11078138.

Experience

Draper Laboratory, Cambridge, MA

September 2025 – Present

Draper Scholar

- Conduct research on AI-enabled inspection systems, with emphasis on multimodal perception, foundation models, uncertainty, and human-in-the-loop decision support.
- Investigate system reliability, model confidence, and deployment constraints for intelligent inspection workflows in safety-critical settings.

University of Massachusetts Lowell, Lowell, MA

December 2023 – December 2024

Robotics Research Assistant in Assisted Navigation

- Managed day-to-day robotics lab operations, including equipment, system setup, and maintenance of research hardware/software environments.
- Conducted research in assistive robotics and multimodal human–robot interaction, including perception, navigation, and vision-language interfaces.
- Conducted participatory design and user evaluation with target users; published the resulting system at IEEE Ubiquitous Robots 2025.

CarGurus, Cambridge, MA

July 2022 – December 2022

Software Engineering Co-op

- Integrated a machine-learning recommendation API into the Search Results Page to support personalized vehicle recommendations.

Research Projects

Weld-PICA — Uncertainty-Aware Vision-Language Weld Inspection

- Developed a teacher–student VLM system for weld inspection with defect classification, explanations, and uncertainty estimation.
- Evaluated reliability using calibration, false accept/reject rates, and human review for resource-constrained deployment.

CROSS-VLR — System-Level Evaluation of Vision-Language Robotics

- Developed a systems-engineering evaluation framework for vision-language robotics, adapting HFACS-style failure analysis to decompose performance across perception, reasoning, and action.
- Applied the framework across assistive navigation, industrial inspection, and human–robot interaction to localize failure sources beyond end-to-end task success.

Seeing Eye Stretch – Python, C++, ROS2, PyTorch, Ultralytics

- Developed a Hello Robot Stretch RE2 assistive navigation system for blind and low-vision users.
- Integrated vision, LiDAR/infrared navigation, and multimodal VQA for environmental understanding and user interaction.

Image Caption Generation using PEFT – Python, PyTorch, Generative AI, Computer Vision, Multimodal

- Fine-tuned multimodal image-captioning models using LoRA, DoRA, and quantized training to explore parameter-efficient adaptation.
- Implemented and compared multimodal transformer architectures, including ViT–GPT-2 and custom transformer configurations.

Skills

Languages: Python, C++, Java, Swift, JavaScript/TypeScript

ML / AI: PyTorch, TensorFlow, Transformers, PEFT/LoRA, Quantization, RAG, Computer Vision

Robotics / Systems: ROS2, LiDAR/RGB-D perception, Linux, Git